

Computer Simulation

Takumi Miyoshi, Taku Yamazaki

Special thanks to: Dr. Pham Thi Huong

Introduction

- Instructor: Takumi Miyoshi
- Education:
 - 1996/04 – 1999/03:** PhD, University of Tokyo
 - 1994/04 – 1996/03:** M.Eng., University of Tokyo
 - 1990/04 – 1994/03:** B.Eng., University of Tokyo
- Professional career:
 - 2001/04 – current:** Shibaura Institute of Technology
 - 1999/04 – 2001/03:** Waseda University
 - 2010/06 – 2011/03:** Sorbonne Université (former UPMC)
- Major:
 - Computer network, P2P network, Smart city

Takumi Miyoshi / Taku Yamazaki

Lesson 1: Introduction

2

Introduction

- Instructor: Taku Yamazaki
- Education:
 - 2014/04 – 2017/03:** PhD, Waseda University
 - 2012/04 – 2014/03:** M.Eng., Shibaura Institute of Technology
 - 2008/04 – 2012/03:** B.Eng., Shibaura Institute of Technology
- Professional career:
 - 2018/04 – current:** Shibaura Institute of Technology
 - 2015/11 – 2018/03:** Waseda University
- Major:
 - Wireless network, Ad hoc network, IoT, Smart city

Takumi Miyoshi / Taku Yamazaki

Lesson 1: Introduction

3

Lesson 1

Introduction to the course

- Introduction
- Course Objectives
- Course Organization
- How to Learn and Evaluation
- Course Materials (official website, books)
- MATLAB (history, applications)
- Examples

Takumi Miyoshi / Taku Yamazaki

Lesson 1: Introduction

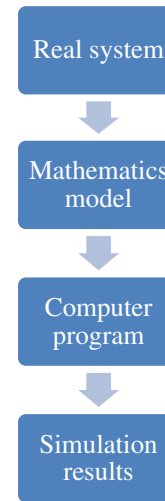
4

What is Computer Simulation?

- Computer simulation is a computer program that reproduces the behavior of a system using a mathematical model⁽¹⁾
- Simulation process
 - Creating a mathematical model represents the physical object (declarative, functional, constraint, spatial or multi model)
 - Executing the model on a computer → create a computer program
- Computer simulation allows:
 - Making predictions based on past events
 - Understanding the system's behaviors
 - Exploratory or heuristic

(1) Wikipedia article in computer simulation

Computer Simulation Process



- Collect the statistical data sets of a real system
- Create an algorithm or a mathematical model that is close with the characteristics of the real system
- Make the program
- Obtain and analyze results

Examples

- Real system: Consider a call center where approximately 60 calls per hour have to be handed. Serving time for a customer is approximately exponentially distributed with the mean of 3 minutes. The question is how many operators are needed to average waiting time is less than 1 minute.
- Mathematics model: queuing system
 - Arrival process according to Poisson with $\lambda = 60$ calls per hour = 1 call per minute
 - Serving time according to exponentially distributed with $\mu = 1/3$ call per minute
 - Average waiting time should be less than 1 minute
 - Find the number of operators
- Make program

Why do we need Computer Simulation?

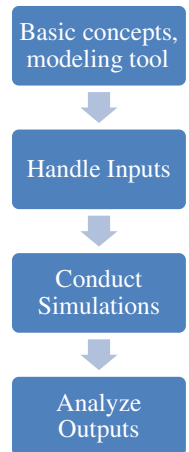
- Safe way to test and explore different scenarios
- Save cost and time
- Visualization: graphics are animated in 2D/3D, easily verified, communicated, and understood
- Allow observation of system behavior any time, at any level of detail
- Debugging: detect errors, represent performance information → allow risk quantification, tuning data

When to use Computer Simulation?

- To study a dynamic behavior in a dangerous environment (difficult to implement in real life)
- To determine behaviors when individual components of a system are altered
- To determine potential effects such as that of river systems for the construction of dams
- To make everything is easier to understand (e.g. the weather forecasting)
- Advantages
 - Solves real-world problems safely and efficiently
 - An analysis method which is easily verified, communicated, and understood

Course Objectives

- Basic theory
 - Understanding the basic concepts of computer simulation and modeling
 - Using a typical simulation tool: MATLAB
- Applications
 - Handling the simulation data
 - Conducting the practical simulations
 - Analyzing the simulation results



Course Organization

- Lesson 1: Introduction to the course
- Lesson 2: How to use MATLAB
- Lesson 3: Flow Controls
- Lesson 4: User-defined Functions
- Lesson 5: Graphics
- Lesson 6: File Operation
- Lesson 7&8&9: Queue Theory
- Lesson 10&11: Slotted-Aloha like Network
- Lesson 12&13: Program Modification and Analysis
- Lesson 14: Final Assignment

Course Material

- Official MATLAB website
<https://ch.mathworks.com/help/matlab/getting-started-with-matlab.html>
- Queuing Systems
L. Kleinrock, "Queueing Systems, Volume 1: Theory," Wiley, 1975
- Lecture notes of Computer Simulation, Bo Gu and Cheng Zhang, 2016-2017
- Lecture notes of Computer Simulation, Pham Thi Huong, 2018-2019
- Stormy Attaway, "MATLAB: a practical approach," 2009
- Others useful resources:
 - <https://ocw.mit.edu/courses/mathematics/18-s997-introduction-to-matlab-programming-fall-2011/>
 - https://en.wikipedia.org/wiki/Computer_simulation

How to Learn and Evaluation

- How to learn this course
 - Review lesson before the class
 - Attend the class
 - Practice as much as possible
 - Do all home assignments
- Evaluation
 - The final grade will be based on your activities
 - Assignments, mini tests, reports and presentations

MATLAB

History and Applications

- What is MATLAB
 - MATLAB is short of Matrix Laboratory
 - Started developing in the late 1970s by Prof. Cleve Moler
 - Initial release in 1984 by MathWorks
 - A high-performance language for numerical computing environment
 - Integrate computation, visualization, and programming environment
- Applications
 - Matrix manipulations, math and computation
 - Implementation of algorithms, modeling, and simulation
 - Plotting functions and data
 - Interfacing with programs written in other languages (C, C++, C#, Java, Python, and Fortran)

MATLAB

Examples

- Examples of Matrix manipulations, math and computation

```
>> A=eye(3)
```

```
A =  
    1    0    0  
    0    1    0  
    0    0    1
```

```
>> b=2*3*4
```

```
b =  
    24
```

MATLAB

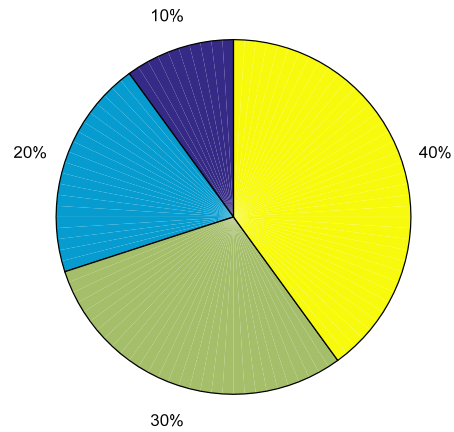
Examples

- Implementation of algorithms, modeling, and simulation → Algorithms to find the roots of a quadratic equation (ptb2)

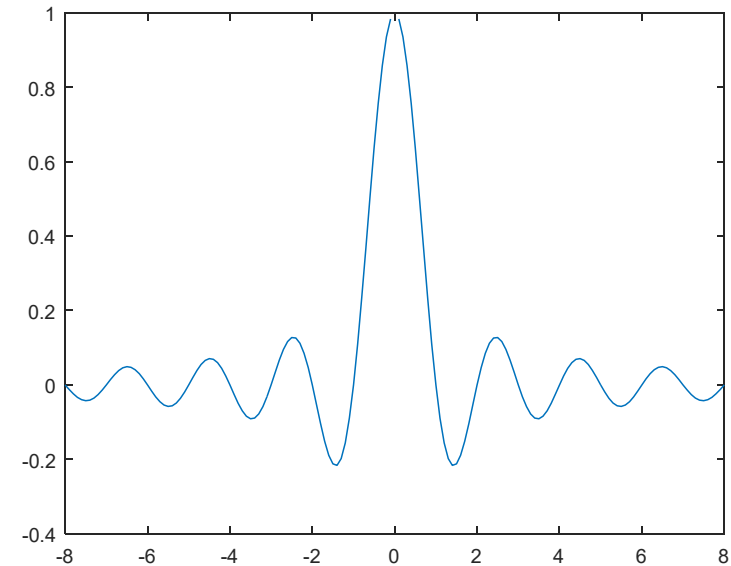
MATLAB

Examples: Plotting

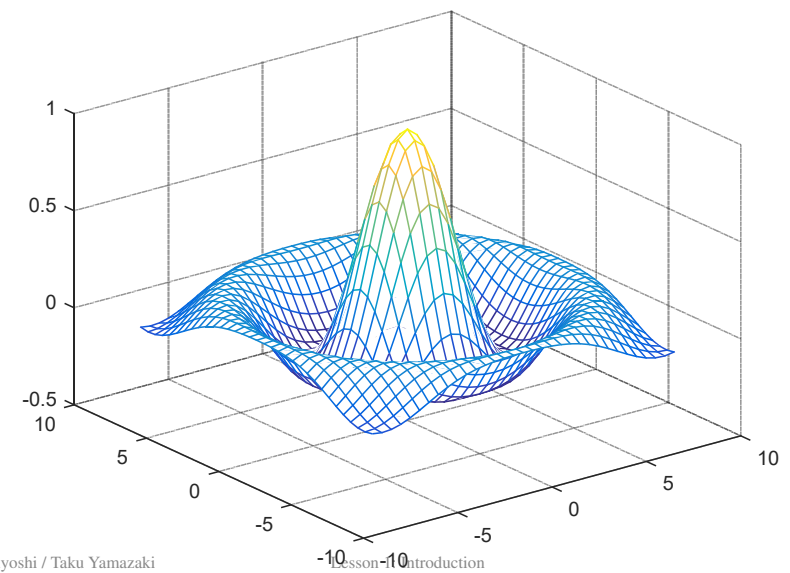
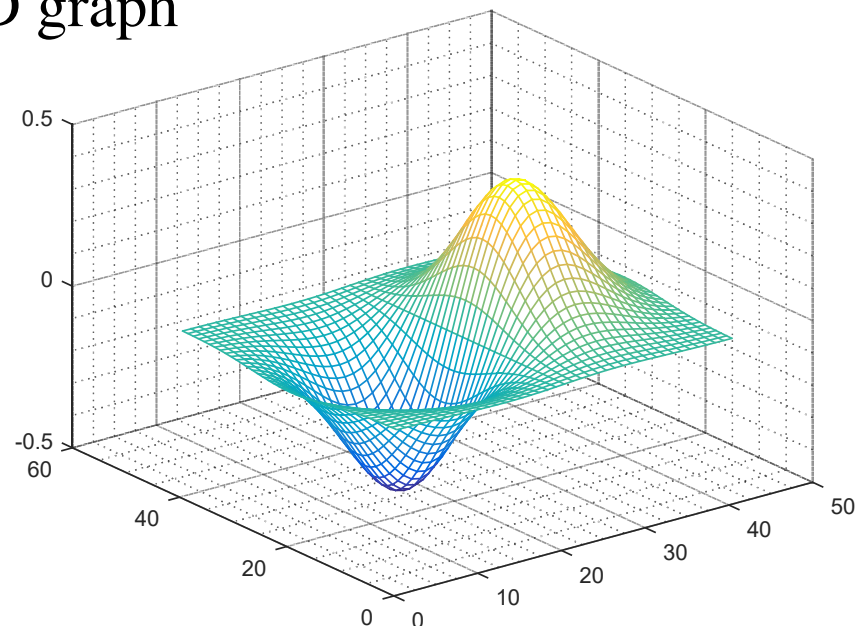
- `pie([5 10 15 20])`



The sinc function



3D graph



Thank you very much for your attention!

MATLAB Installation Guide (1)

Install Manual: https://web.sic.shibaura-it.ac.jp/page_20191024062404

1) Login to MathWorks MATLAB portal

- Go to SIT portal site of MATLAB
 - URL: <https://jp.mathworks.com/academia/tah-portal/shibaura-institute-of-technology-584141.html>
- Go to "Sign in" page
- Login with Gakunin user name and password

2) Make MathWorks account

- If you do not have MathWorks account, create it just after login.
- You must use "SIT mail address" (ex. z120001@shibaura-it.ac.jp) as your user name.

MATLAB Installation Guide (2)

3) Download the installer of MATLAB

- Select and click the current version (R2025b Update 5)
- You will receive the installer

4) Run the installer

- Run the installation program after the download
- MATLAB consumes about 4GB disk space without any option

I hope you can smoothly install and use the software!